# Problem 9.14

Fund $A$ earns interest at a nominal rate of 6% compounded monthly. Fund $B$ earns interest at a nominal rate of discount compounded three times per year. The annual effective rates of interest earned by both funds are equivalent. Calculate the nominal rate of discount earned by Fund $B$.

# Solution

For Fund $A$, the nominal rate compounded monthly is

$$i^{(12)} = 6\%.$$

Thus, the monthly effective rate is

$$\frac{i^{(12)}}{12} = \frac{0.06}{12} = 0.005.$$

For Fund $B$, the nominal discount rate compounded three times per year is $d^{(3)}$. Since the annual effective rates are equivalent, we have

$$\left(1 - \frac{d^{(3)}}{3}\right)^{-3} = \left(1 + \frac{i^{(12)}}{12}\right)^{12}.$$

Substituting $i^{(12)}/12 = 0.005$ gives

$$\left(1 - \frac{d^{(3)}}{3}\right)^{-3} = (1.005)^{12}.$$

Taking both sides to the power $-1/3$,

$$1 - \frac{d^{(3)}}{3} = (1.005)^{-4}.$$

Therefore,

$$d^{(3)} = 3\left(1 - (1.005)^{-4}\right).$$

Finally,

$$d^{(3)} \approx 0.0592574.$$

Hence, the nominal rate of discount earned by Fund $B$ is

$$\boxed{5.92574\%}.$$